Study of Aldosterone Synthase Inhibition as an Add-On Therapy in Resistant Hypertension

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Aldosterone inhibition with mineralcorticoid receptor antagonists (MRAs) is an effective treatment for resistant hypertension. Aldosterone synthase inhibitors (ASIs) are currently being investigated as a new therapeutic strategy to reduce aldosterone secretion from the adrenal gland. In this study, the efficacy and safety of the first-generation ASI LCI699 (0.25 mg twice daily, 1 mg 4 once daily, and 0.5 mg/1 mg twice daily) was compared with placebo and eplerenone (50 mg twice daily), in patients with resistant hypertension. Placebo-adjusted decreases in systolic blood pressure (BP) with LCI699 ranged from 2.6 mm Hg to 4.3 mm Hg at week 8; changes in diastolic BP ranged from +0.3 mm Hg to −1.2 mm Hg. However, reductions were smaller than observed with eplerenone 50 mg twice daily (9.9 mm Hg and 2.9 mm Hg for systolic and diastolic BP, respectively) and not statistically significant vs placebo. LCI699 suppressed plasma aldosterone levels in a dose-related manner with corresponding dose-dependent increases in plasma renin activity and plasma 11-deoxycorticosterone. LCI699 and eplerenone were well tolerated. These data demonstrate that aldosterone synthesis inhibition with LCI699 lowers BP modestly in patients with resistant hypertension. Aldosterone synthesis inhibition might offer an attractive adjunct to aldosterone receptor blockade, although the potential of a combination MRA/ASI has not yet been tested. *J Clin Hypertens (Greenwich)*. 2013; 15:186–192. ©2012 Wiley Periodicals, Inc.

Aldosterone plays an important role in blood pressure (BP) control and maintaining potassium and sodium homeostasis and in the pathophysiology of cardiovascular and renal disease.[1–4] A relative aldosterone excess predicts hypertension onset[5] and aldosterone is believed to play a critical role in mediating and aggravating resistant hypertension,[6] a common clinical problem that affects 20% to 28% of patients with hypertension.[7,8] It was previously thought that primary aldosteronism is present in <1% of unselected hypertensive patients, but more recent estimates suggest that it affects closer to 10%, and these patients tend to have higher incidences of cardiovascular (CV) damage when controlled for sex, age, and BP status.[9] Thus, aldosterone blockade represents an attractive therapeutic approach to lower BP and to reduce the risk of CV events and end-organ damage.

One approach for blocking the effects of aldosterone is to prevent its binding to mineralocorticoid receptors, and two mineralocorticoid receptor antagonists (MRAs), eplerenone and spironolactone, have been shown to be effective for the treatment of hypertension and heart failure.[10–14] A reduction in BP with spironolactone was associated with reduction in cardiac mass and improvement in diastolic function in patients with resistant hypertension.[15] However, MRAs induce a counter-regulatory increase in aldosterone concentration, which may limit the efficacy of mineralocorticoid receptor (MR) blockade by stimulating MR-independent effects of aldosterone.[16] Indeed, aldosterone is known to have both genomic effects, mediated via activation of the MR, and rapid nongenomic effects regulating the contractility of the CV system.[17] Thus, an alternative approach to attenuate the effects of aldosterone would be to decrease circulating levels by inhibiting its synthesis, preferably at the aldosterone synthase enzyme (CYP11B2), which mediates the rate-limiting conversion of 11-deoxycorticosterone (11-DOC) to aldosterone.

LCI699 is a potent, orally active aldosterone synthase inhibitor (ASI) that has been developed for human use. In a clinical study in sodium-depleted healthy men, LCI699 demonstrated significant dose-dependent suppression of plasma and urine aldosterone levels compared with placebo and was well tolerated at doses up to 3.0 mg/d following oral administration over 14 days.[18] Because LCI699 has been shown to effectively correct hypokalemia in patients with primary aldosteronism and to dose-dependently lower BP in patients with essential hypertension,[19,20] a study in difficult-to-treat groups, such as patients with resistant hypertension was initiated. The present study explored the effects of a range of LCI699 doses as add-on therapy to an ongoing 3-drug regimen in patients with resistant hypertension.

METHODS

Study Design

This was a prospective, randomized, double-blind, placebo- and active-controlled, parallel-group, multicenter,

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dose-ranging study to compare the safety and efficacy of LCI699 in patients with resistant hypertension conducted at 37 centers in the United States and 1 center in Iceland. The study was conducted according to the ethical principles of the Declaration of Helsinki and was approved by the independent ethics committee or institutional review board at each center. Informed, written consent was obtained from each participant before randomization. The trial is registered with ClinicalTrials.gov, identifier NCT00817635.

After a 2-week single-blind placebo run-in, patients were randomized (1:1:1:1:1) to receive placebo, eplerenone 50 mg twice daily (BID) or 1 of 3 LCI699 treatment groups (LCI699 0.25 mg BID, 1 mg once daily [QD], 0.5 mg BID titrated to 1 mg BID after 4 weeks) for 8 weeks (Figure 1). Patients received their study medications in morning bottles (to be administered at approximately 9 AM) and evening bottles (to be administered at approximately 9 PM) and were instructed to swallow the capsules whole. All patients were treated for 8 weeks. Eplerenone was selected as the active comparator due to its similar pharmacokinetic properties to LCI699 and recommended BID dosing for hypertension. A follow-up safety visit was conducted 2 weeks after completing the treatment.

Patients

Men and women aged 18 to 75 years with resistant hypertension prior to randomization were considered for this trial. Resistant hypertension was defined as failure to achieve BP goals in patients adhering to optimal doses of a stable $\geq$3 drug antihypertensive regimen that included a diuretic for a period of at least 4 weeks. Optimal doses were defined as: the highest dose listed by the Seventh Report of the Joint National Committee on Prevention, Detection, Evaluation, and Treatment of High Blood Pressure (JNC7) guidelines, the highest allowable prescribed dose as per the label, the highest dose tolerated by an individual patient, or the highest dose appropriate for a patient in the judgment of the investigator. Concomitant antihypertensive medications were continued for the duration of the study. All prior nonstudy medications were allowed (unless specifically contraindicated in the exclusion criteria), provided that the patient was on a standard treatment regimen and there was no intent to change treatment during the study.

The main study exclusion criteria included recent history (within past 6 months) of myocardial infarction, heart failure, unstable angina, coronary artery bypass graft, percutaneous coronary intervention, hypertensive encephalopathy, cerebrovascular accident, or transient ischemic attack, clinically significant cardiac conduction defects, clinically significant valvular heart disease, pregnant or nursing women, a history of secondary hypertension of any etiology, type 1 diabetes or type 2 diabetes mellitus with poor glucose control (hemoglobin A_{1c} >9%), patients with creatinine clearance <50 mL/min (by the Modification of Diet in Renal Disease formula), and patients taking aldosterone receptor antagonists, direct renin antagonists or potassium-sparing diuretics within 4 weeks of screening.

Objectives

The primary objective of the study was to explore the effect of various LCI699 regimens on mean sitting systolic BP (msSBP) compared with placebo following 8 weeks of treatment. The secondary objectives included similar monitoring of mean sitting diastolic BP (msDBP), exploring changes from baseline in mean 24-hour, daytime, and nighttime systolic BP (SBP) and diastolic BP (DBP) as assessed by 24-hour ambulatory BP monitoring (ABPM). Correlations between the BP-lowering effect of LCI699 and changes in renin-angiotensin-aldosterone system (RAAS) biomarkers (plasma aldosterone level, plasma aldosterone to renin ratio, and plasma renin activity [PRA]) in response to treatment were evaluated as exploratory objectives.

Assessments

Office BP and heart rate was measured throughout the study at every visit except for the randomization visit. BP was measured using an automated device (such as the Omron BP monitor) in accordance with British Hypertension Society 2004 guidelines.[18] Office BP and heart rate measurements were performed at trough concentration (23–26 hours post-morning dose) during

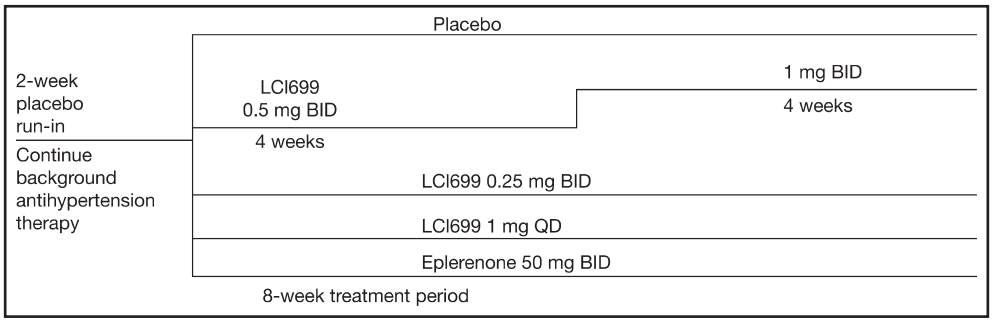

FIGURE 1. Study design. BID indicates twice daily; QD, once daily.

TABLE I. Baseline Demographics and Characteristics by Treatment Group (Full Analysis Set)

Demographic variable	LCI699 0.25 mg BID (n=32)	LCI699 1 mg QD (n=26)	LCI699 0.5/1 mg BID (n=31)	Eplerenone 50 mg BID (n=33)	Placebo (n=33)	Overall Total (N=155)
Age, mean (SD), y	53.6 (10.36)	55.4 (9.58)	57.2 (10.77)	56.2 (7.70)	59.8 (9.33)	56.5 (9.69)
Age group, No. (%), y						
<65	28 (87.5)	23 (88.5)	23 (74.2)	30 (90.9)	20 (60.6)	124 (80.0)
≥65	4 (12.5)	3 (11.5)	8 (25.8)	3 (9.1)	13 (39.4)	31 (20.0)
Sex, No. (%)						
Male	20 (62.5)	18 (69.2)	18 (58.1)	19 (57.6)	22 (66.7)	97 (62.6)
Female	12 (37.5)	8 (30.8)	13 (41.9)	14 (42.4)	11 (33.3)	58 (37.4)
Race, No. (%)						
Caucasian	21 (65.6)	15 (57.7)	21 (67.7)	18 (54.5)	22 (66.7)	97 (62.6)
Black	11 (34.4)	9 (34.6)	10 (32.3)	14 (42.4)	11 (33.3)	55 (35.5)
Asian	0	0	0	0	0	0
Other	0	2 (7.7)	0	1 (3.0)	0	3 (1.9)
Baseline BMI, mean (SD), kg/m^2	33.95 (5.918)	32.18 (5.286)	33.49 (8.128)	33.23 (5.519)	31.92 (7.785)	32.98 (6.632)
Duration of hypertension, mean (SD), y	14.1 (10.33)	11.7 (9.34)	13.9 (9.71)	13.0 (8.33)	15.5 (9.64)	13.7 (9.44)
Antihypertensive medications, No. (%)						
3	22 (68.8)	20 (76.9)	23 (74.2)	25 (75.8)	26 (78.8)	116 (74.8)
4	10 (31.3)	5 (19.2)	7 (22.6)	7 (21.2)	4 (12.1)	33 (21.3)
5	0	1 (3.8)	0	1 (3.0)	2 (6.1)	4 (2.6)
6	0	0	1 (3.2)	0	1 (3.0)	2 (1.3)
Baseline GFR, mean (SD), mL/min/SA	83.5 (16.64)	83.6 (20.77)	79.7 (14.80)	81.8 (14.49)	76.9 (16.65)	81.0 (16.62)
History of diabetes, No. (%)	7 (21.9)	7 (26.9)	8 (25.8)	9 (27.3)	6 (18.2)	37 (23.9)
msSBP, mean (SD), mm Hg	152.4 (11.21)	152.5 (9.79)	152.2 (7.58)	153.8 (8.92)	153.4 (9.61)	152.9 (9.39)
msDBP, mean (SD), mm Hg	91.8 (11.68)	89.2 (9.56)	88.9 (11.89)	89.1 (9.84)	90.1 (11.65)	89.8 (10.92)

Abbreviations: BID, twice daily; BMI, body mass index; GFR, glomerular filtration rate; msDBP, mean sitting diastolic blood pressure; msSBP, mean sitting systolic blood pressure; SA, body surface area; SD, standard deviation; QD, once daily.

TABLE II. Concomitant Antihypertensive Medications Used by Drug Class and Treatment Group (Safety Set)

Drug class, No. (%)	LCI699 0.25 mg BID (n=32)	LCI699 1 mg QD (n=26)	LCI699 0.5/1 mg BID (n=31)	Eplerenone 50 mg BID (n=33)	Placebo (n=33)	Overall Total (N=155)
Hydrochlorothiazide	26 (81.3)	23 (88.5)	28 (90.3)	32 (97.0)	33 (100.0)	142 (91.6)
ARB	23 (71.9)	15 (57.7)	11 (35.5)	21 (63.6)	19 (57.6)	89 (57.4)
Calcium channel blocker	19 (59.4)	12 (46.2)	17 (54.8)	17 (51.5)	21 (63.6)	86 (55.5)
ACE inhibitor	14 (43.8)	14 (53.8)	18 (58.1)	15 (45.5)	17 (51.5)	78 (50.3)
β-Blocker	12 (37.5)	16 (61.5)	18 (58.1)	15 (45.5)	9 (27.3)	70 (45.2)

Abbreviations: ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; BID, twice daily; QD, once daily.

the treatment phase of the study. Standing BP measurements were used to characterize changes in orthostatic BP. During the screening visit, BP was measured in both arms and the arm with the higher SBP reading was used at all subsequent visits. Where possible, each patient had BP measurements obtained by the same staff member, at the same time of day, and using the same equipment at all visits.

Twenty-four–hour ABPM was performed at the completion of the single-blind run-in period after 4 weeks and 8 weeks of double-blind treatment. The ABPM device was attached to the nondominant arm of the patient and removed only after a minimum of 24 hours. ABPM data were evaluated against a set of quality-control criteria and if the criteria were not met, the procedure was repeated, preferably within 48 hours. If a patient did not successfully complete the ABPM measurement after two attempts, the test was considered a failure.

RAAS biomarkers including plasma aldosterone concentration, 11-DOC, PRA, active renin concentration (ARC), and aldosterone/PRA ratio (ARR) were measured at the randomization visit, week 4, week 8, and the follow-up visit of week 10. Blood samples were collected in the morning between 7 AM and 10 AM, following an overnight fast and 15-minute seated rest period, and subjected to biochemical analysis at a central laboratory.

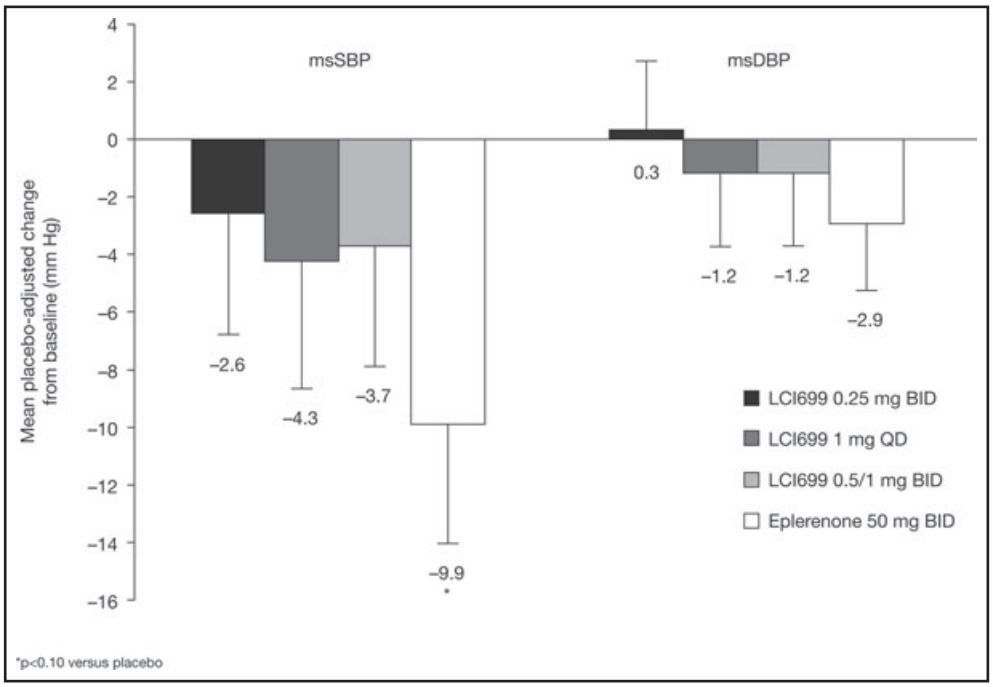

FIGURE 2. Between-treatment analysis for change from baseline in mean sitting systolic blood pressure (msSBP) and mean sitting diastolic blood pressure (msDBP) at week 8 last observation carried forward. BID indicates twice daily; QD, once daily.

Safety assessments included standard adverse event (AE) reporting, and assessments of vital signs, clinical blood chemistry, and hormonal profile. Serum electrolytes were monitored throughout the study.

Statistical Analysis

The primary endpoint was analyzed using a two-way analysis of covariance (ANCOVA) model with baseline msSBP as the continuous covariate with treatment and country as class effects. Each of the three LCI699 dose regimens were compared against placebo, generating 3 two-sided *P* values tested at the 10% level of significance. The 90% confidence interval (CI) of the mean difference (LCI699 vs placebo) is presented for each LCI699 dose. A comparison between the 0.5 mg BID and 1.0 mg QD dose of LCI699 was also undertaken to explore potential regimen effects. The mean change from baseline in msSBP, ABPM measurements, and mean changes from baseline in (log-transformed) RAAS biomarkers were analyzed using the same 2-way ANCOVA model used for the primary endpoint using the last observation carried forward.

RESULTS

Baseline Characteristics

A total of 155 patients (LCI699 0.25 mg BID [n=32], LCI699 1 mg QD [n=26], LCI699 0.5 mg/L mg BID [n=31], eplerenone 50 mg BID [n=33], placebo [n=33]) were randomized for the study and treated with study medication. Twenty-nine patients discontinued the study, with a protocol deviation or withdrawal of informed consent being the most common causes of discontinuation (n=20). Patient demographics and baseline disease characteristics were comparable across the 5 treatment groups (Table I). Across all treatment groups the mean baseline msSBP was 152.9 mm Hg and mean msDBP was 89.8 mm Hg. Antihypertensive medication use is reported in Table II.

Efficacy

After 8 weeks of treatment, all LCI699 groups showed numerical reductions in msSBP but none of these reductions achieved statistical significance vs placebo. Placebo-adjusted msSBP reductions were numerically larger with LCI699 1 mg QD and 0.5/1 mg BID (4.3 mm compared with LCI699 0.25 mg BID group) (Figure 2). The largest mean placebo-adjusted reduction in msSBP was observed in the eplerenone 50 mg BID group (9.9 mm Hg; *P*=.017 vs placebo). LCI699 and eplerenone produced modest placebo-adjusted changes from baseline in msDBP, although none were statistically significantly different from placebo (Figure 2).

Mean changes from baseline in 24-hour mean ambulatory SBP (mASBP), mean ambulatory DBP (mADBP), mean daytime SBP/DBP, and mean nighttime SBP/DBP followed a pattern similar to changes in msSBP (Figure 3). The largest placebo-adjusted mean reductions in ambulatory BP parameters at week 8 were observed in the eplerenone group (14.7 mm Hg [mASBP], 14.1 mm Hg [daytime mean

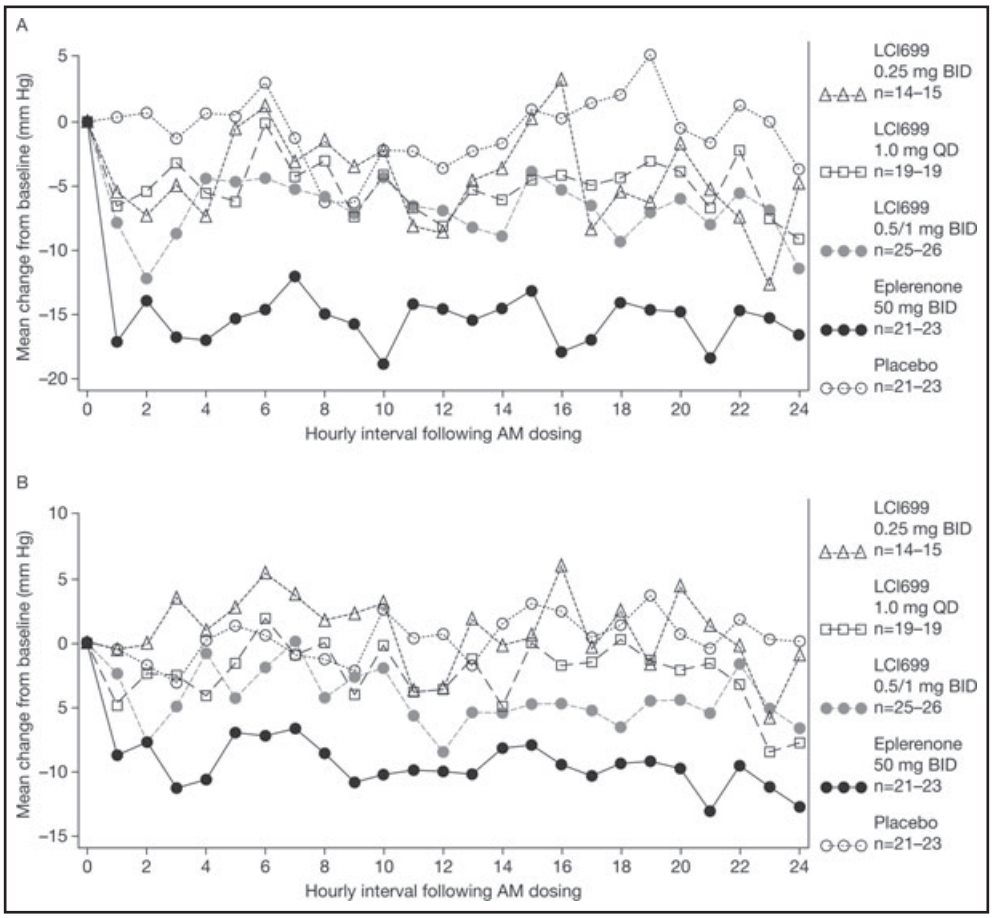

FIGURE 3. Mean change from baseline to week 8 (last observation carried forward) in hourly average systolic blood pressure (A) and diastolic blood pressure (B) as measured by ambulatory blood pressure monitoring (full analysis set). BID indicates twice daily; QD, once daily.

TABLE III. Effect of LCI699 and Eplerenone on RAAS Biomarkers

	LCI699 0.25 mg BID (n=32)	LCI699 1 mg QD (n=26)	LCI699 0.5/1 mg BID (n=31)	Eplerenone 50 mg BID (n=33)
Placebo-adjusted change from baseline at week 8 (%) (LOCF)				
Plasma aldosterone	−18.2	−26.8	−50.7[a]	126.2[b]
PRA	44.7	78.1	112.2[c]	425.3[b]
ARC	37.0	36.8	103.0[c]	320.0[b]
11-DOC	90.5[d]	106.3[d]	375.2[d]	6.2
ARR	−43.8	−47.3	−77.1[b]	−54.8[c]

Abbreviations: ARC, active renin concentration; ARR, aldosterone/PRA ratio; BID, twice daily; 11-DOC, plasma 11-deoxycorticosterone; LOCF, last observation carried forward; PRA, plasma renin activity; RAAS, renin-angiotensin-aldosterone system; QD, once daily. [a]$P<.10$. [b]$P<.001$ vs placebo. [c]$P<.01$ vs placebo. [d]$P<.05$ vs placebo.

SBP], 15.8 mm Hg [nighttime mean SBP] ($P<.001$ for all) and 9.4 mm Hg (mADBP), 8.8 mm Hg (daytime mean DBP), 10.8 mm Hg (nighttime mean DBP) ($P\leq.001$ for all). All of the above were not statistically significant in the LCI699 treatment groups except for the week 8 reduction in nighttime mean SBP and DBP of 7.4 mm Hg ($P=.060$) and 5.8 mm Hg ($P=.020$), respectively, in the LCI699 0.5/1 mg BID group.

Changes in RAAS biomarkers at week 8 are outlined in Table III. LCI699 suppressed plasma aldosterone and increased PRA in a dose-dependent manner

compared with placebo. Eplerenone significantly increased both plasma aldosterone and PRA. Both LCI699 and eplerenone increased ARC. LCI699, but not eplerenone, significantly increased 11-DOC. Eplerenone, despite increasing plasma aldosterone levels, significantly decreased ARR.

Safety and Tolerability

No deaths were reported and there was only one serious AE (in a patient randomized to eplerenone). Six patients discontinued because of treatment-emergent AEs or abnormal laboratory values (LCI699 0.25 mg BID [n=2], 1 mg QD [n=1]; eplerenone [n=1]; placebo [n=2]). The number of patients experiencing an AE was similar among the LCI699 (42.7%), eplerenone (39.4%), and placebo (48.5%) groups. The majority of AEs were mild and not considered related to the study drug. Hyponatremia was reported more frequently with LCI699 and eplerenone treatment (5.6% and 6.1%, respectively) than with placebo (0%).

Small but statistically significant increases in serum potassium were observed with both LCI699 and eplerenone (placebo-adjusted mean increases from baseline of 0.1 to 0.4 mmol/L; $P<.1$). Statistically significant decreases in serum sodium from baseline were also observed in the eplerenone and LCI699 0.5/1.0 mg BID groups (0.9–2.1 mmol/L; $P<.1$). Changes in serum potassium and sodium returned toward baseline values at the follow-up visit.

DISCUSSION

In the present study, numerical decreases in msSBP and msDBP were observed among LCI699-treated patients with resistant hypertension. However, these reductions were smaller than those observed with eplerenone 50 mg BID, and placebo-adjusted changes from baseline were not statistically significant. In contrast, a number of studies with MRAs as fourth-line antihypertensive therapy for uncontrolled BP in patients with resistant hypertension have reported large, significant reductions in SBP and DBP with these treatments.[21,22] In addition, LCI699 has previously been reported to significantly reduce BP in patients with essential hypertension, with the 1-mg QD dosage reducing BP to a similar extent to eplerenone 50 mg.[20]

The reasons for the lack of BP-lowering efficacy in resistant hypertension compared with essential hypertension are unclear, as is the apparent inferiority of LCI699 in direct comparison to eplerenone in this study. It may be that higher LCI699 doses are necessary to reduce BP similar to eplerenone in these patients. Unfortunately, higher doses of LCI699 are unlikely to be tested due to concern over off-target effects on cortisol synthesis due to loss of steroidogenic target selectivity.[18,20,23] A potential explanation may be that non-aldosterone compounds, such as 11-DOC and cortisol, play a larger role in activating MR irrespective of the circulating levels of aldosterone. Indeed, while the recognition of the role of aldosterone in BP and BP-related CV damage has clearly expanded, it is likely that the MR receptor plays more of a major role in essential hypertension, cardiac failure, and atherosclerosis than the hormone.[9]

CONCLUSIONS

In the present study, eplerenone increased plasma aldosterone concentrations, which is consistent with previous MRA treatment. Since aldosterone may have nongenomic effects mediated independently of mineralocorticoid receptors,[17] and elevated aldosterone levels observed with use of MRAs have been implicated in mediating worse glucose homeostasis and increased central sympathetic drive,[24,25] inhibition of aldosterone synthesis might offer an attractive adjunct to aldosterone receptor blockade by MRAs.[26] Although the current study shows that LCI699 at the doses tested is able to lower BP modestly while reducing circulating aldosterone levels in patients with resistant hypertension, further research is needed to elucidate the clinical benefits and limitations of an ASI added to MRAs. It has yet to be tested whether a combination MRA/ASI may be more effective than MRA treatment alone.

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References

1. Brown NJ. Aldosterone and end-organ damage. *Curr Opin Nephrol Hypertens* 2005;14:235–241.
2. Cohn JN, Colucci W. Cardiovascular effects of aldosterone and post-acute myocardial infarction pathophysiology. *Am J Cardiol* 2006;97:4F–12F.
3. Pimenta E, Calhoun DA. Aldosterone, dietary salt, and renal disease. *Hypertension* 2006;48:209–210.
4. Funder JW, Mihailidou AS. Aldosterone and mineralocorticoid receptors: clinical studies and basic biology. *Mol Cell Endocrinol* 2009;301:2–6.
5. Vasan RS, Evans JC, Larson MG, et al. Serum aldosterone and the incidence of hypertension in nonhypertensive persons. *N Engl J Med* 2004;351:33–41.
6. Whaley-Connell A, Johnson MS, Sowers JR. Aldosterone: role in the cardiometabolic syndrome and resistant hypertension. *Prog Cardiovasc Dis* 2010;52:401–409.
7. Calhoun DA, Jones D, Textor S, et al. Resistant hypertension: diagnosis, evaluation, and treatment: a scientific statement from the American Heart Association Professional Education Committee of the Council for High Blood Pressure Research. *Hypertension* 2008;51:1403–1419.
8. Egan BM, Zhao Y, Axon RN, et al. Uncontrolled and apparent treatment resistant hypertension in the United States, 1988 to 2008. *Circulation* 2011;124:1046–1058.
9. Funder JW. The role of aldosterone and mineralocorticoid receptors in cardiovascular disease. *Am J Cardiovasc Drugs* 2007;7:151–157.
10. Pitt B, Zannad F, Remme WJ, et al. The effect of spironolactone on morbidity and mortality in patients with severe heart failure. Randomized Aldactone Evaluation Study Investigators. *N Engl J Med* 1999;341:709–717.

11. Pitt B, Remme W, Zannad F, et al; Eplerenone Post-Acute Myocardial Infarction Heart Failure Efficacy and Survival Study Investigators. Eplerenone, a selective aldosterone blocker, in patients with left ventricular dysfunction after myocardial infarction. *N Engl J Med* 2003;348:1309–1321.
12. Zannad F, McMurray JJ, Krum H, et al. Eplerenone in patients with systolic heart failure and mild symptoms. *N Engl J Med* 2011;364:11–21.
13. Václavík J, Sedlák R, Plachy M, et al. Addition of spironolactone in patients with resistant arterial hypertension (ASPIRANT): a randomized, double-blind, placebo-controlled trial. *Hypertension* 2011;57:1069–1075.
14. Williams GH, Burgess E, Kolloch RE, et al. Efficacy of eplerenone versus enalapril as monotherapy in systemic hypertension. *Am J Cardiol* 2004;93:990–996.
15. Ubaid-Girioli S, Adriana de Souza L, Yugar-Toledo JC, et al. Aldosterone excess or escape: treating resistant hypertension. *J Clin Hypertens* 2009;11:245–252.
16. Brown R, Quirk J, Kirkpatrick P. Eplerenone. *Nat Rev Drug Discov* 2003;2:177–178.
17. Schriffrin EL. Effects of aldosterone on the vasculature. *Hypertension* 2006;47:312–318.
18. Menard J, Watson C, Rebello S, et al. Hormonal and electrolyte responses to the aldosterone synthase inhibitor LCI699 in sodium depleted healthy subjects. *J Am Coll Cardiol* 2010;55:A61.E581.
19. Amar L, Azizi M, Menard J, et al. Aldosterone synthase inhibition with LCI699: a proof-of-concept study in patients with primary aldosteronism. *Hypertension* 2010;56:831–838.
20. Calhoun DA, White WB, Krum H, et al. Effects of a novel aldosterone synthase inhibitor for treatment of primary hypertension: results of a randomized, double-blind, placebo- and active-controlled phase 2 trial. *Circulation* 2011;124:1945–1955.
21. Chapman N, Dobson J, Wilson S, et al. Anglo-Scandinavian Cardiac Outcomes Trial Investigators. Effect of spironolactone on blood pressure in subjects with resistant hypertension. *Hypertension* 2007;49:839–845.
22. Calhoun DA, White WB. Effectiveness of the selective aldosterone blocker, eplerenone, in patients with resistant hypertension. *J Am Soc Hypertens* 2008;2:462–468.
23. Andersen K, Hartman D, Peppard T, et al. The effects of aldosterone synthase inhibition on aldosterone and cortisol in patients with hypertension: a phase II, randomized, double-blind, placebo-controlled, multicenter study. *J Clin Hypertens* 2012;14:580–587.
24. Stas S, Whaley-Connell AT, Sowers JR. Aldosterone and hypertension in the cardiometabolic syndrome. *J Clin Hypertens (Greenwich)* 2008;10:94–96.
25. Yu Y, Wei SG, Zhang ZH, et al. Does aldosterone upregulate the brain renin-angiotensin system in rats with heart failure? *Hypertension* 2008;51:727–733.
26. Hollenberg NK. Aldosterone synthase inhibition: a promising beginning. *Curr Hypertens Rep* 2011;13:99–102.